

Algebraic manipulation



- 1
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|--------------------------------|--------------------------------|---|----------------------------|
| a $x = 5, -5$ | b $x = \pm 7\sqrt{2}$ | c $x = \frac{8}{9}, \frac{-8}{9}$ | d $p = \pm 12$ |
| e $x = 0, \frac{12}{7}$ | f $x = 4 \pm \sqrt{10}$ | g $x = \frac{5}{4}, -\frac{11}{4}$ | h $x = \frac{9}{4}$ |
| i $x = 1$ | j $m = 0, 10$ | | |

2 $x = \pm \sqrt{\frac{c}{a}}$

Factorisation

- 3
- | | | | |
|---------------------------------|--|-------------------------------|--------------------------------|
| a $x = 0, -12$ | b $y = 0, \frac{1}{5}$ | c $m = 0, 14$ | d $x = -10, -9$ |
| e $m = 5, -4$ | f $x = 10$ | g $x = -4, 5$ | h $m = 14, 13$ |
| i $x = 8, -3$ | j $y = -1, 3$ | k $x = 2, -4$ | l $x = \frac{9}{4}, 4$ |
| m $x = -\frac{5}{4}, -3$ | n $x = -\frac{4}{5}, -2$ | o $x = \frac{5}{4}, 3$ | p $x = -\frac{5}{3}, 4$ |
| q $x = \frac{-5}{2}, 12$ | r $b = -\frac{5}{3}, \frac{3}{4}$ | s $k = 1, \frac{7}{6}$ | |

4 $x = \pm 4$

5 $x = 2$

6

a $(x - 7)(x + 4) = 0$	b $x(x + 1) = 0$
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7 $x^2 + 8x + 15 = 0$

Using technology

- 8
- | | | |
|--------------------------------|--|--------------------------|
| a $x = 2, -\frac{5}{3}$ | b $x = \frac{3}{2}, -\frac{7}{2}$ | c $x = 0.4, -2.0$ |
|--------------------------------|--|--------------------------|

9

a $x = \pm\sqrt{2}$



b $x = 1.4, -1.4$

10

a 1

b 1

Applications

11 None

12 $x = \frac{12 \pm \sqrt{1044}}{90}$

13 $x = -\frac{3}{5}, -2, -\frac{5}{2}, -4$

14 $k = -16$

15 $a = 13$

16 $t = 22, 14$

17

a $t = 4, 8$

b 4 hours

c 4 hours

18

a $x = 0, 32$

b 32 cm

19

a $17 + \frac{6426}{x^2} = \frac{663}{x}$

b $x = 18, 21$

c 36.83 cm

20

a 0, 14 months

b No, after one year the company will still be spending more money than it is making.